

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 38 | Date: 12-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (4 Questions - 8 Marks)

Q1. [2 marks] (Easy)

Which of the following is an example of a linear data structure?

Options:

A) Tree B) Graph C) Stack D) Hash Table

Q4. [2 marks] (Medium)

Which data structure is best suited for implementing a "undo" functionality in a text editor?

Options:

A) Queue B) Array C) Stack D) Linked List

Q5. [2 marks] (Medium)

Given an empty stack, what will be the state of the stack after the following operations: Push(10), Push(20), Pop(), Push(30), Pop()?

Options:

A) [10] B) [30] C) [10, 30] D) Empty

Q8. [2 marks] (Hard)

What is the worst-case time complexity for searching an element in a Binary Search Tree (BST) that is completely skewed (e.g., all nodes are inserted in ascending order)?

Options:

A) $O(1)$ B) $O(\log n)$ C) $O(n)$ D) $O(n \log n)$

Section 2: Short Answer (6 Questions - 30 Marks)

Q2. [5 marks] (Easy)

Define an Abstract Data Type (ADT) and explain why it is important in data structures.

Q3. [5 marks] (Easy)

Distinguish between an array and a linked list based on their memory allocation strategy.

Q6. [5 marks] (Medium)

Explain the "front" and "rear" pointers in a simple queue and describe their movement during enqueue and dequeue

operations.

Q7. [5 marks] (Medium)

Describe the main advantage of a doubly linked list over a singly linked list. Provide an example scenario where this advantage is particularly useful.

Q9. [5 marks] (Hard)

Describe the concept of hashing and explain how a hash function maps keys to indices in a hash table.

Q10. [5 marks] (Hard)

Compare the average time complexities for searching an element in an unsorted array versus a sorted array using linear search and binary search.

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